

Session	Module	Title
1	Virtual Inaugural	Virtual Inaugural
2	Module 1: Programming & Data Foundations	Introduction to Python for Data Science
3	Module 1: Programming & Data Foundations	Control Flow & Logic
4	Module 1: Programming & Data Foundations	Iterative Programming & Loops
5	Module 1: Programming & Data Foundations	Modular Code with Functions
6	Module 1: Programming & Data Foundations	Core Data Structures
7	Module 1: Programming & Data Foundations	The Notebook Workflow
8	Module 1: Programming & Data Foundations	Working with Structured Data (CSV)
9	Module 1: Programming & Data Foundations	Handling Semi-Structured Data (JSON)
10	Module 1: Programming & Data Foundations	Data Validation & Schema Checks
11	Module 1: Programming & Data Foundations	API Fundamentals & REST
12	Module 1: Programming & Data Foundations	Managing API Complexity
13	Module 1: Programming & Data Foundations	End-to-End Data Fetching Pipeline
14	Module 2: Data Analysis with NumPy & Pandas	NumPy Arrays & Performance
15	Module 2: Data Analysis with NumPy & Pandas	Vectorized Operations
16	Module 2: Data Analysis with NumPy & Pandas	Numeric Manipulation & Linear Algebra Basics
17	Module 2: Data Analysis with NumPy & Pandas	Introduction to Pandas DataFrames
18	Module 2: Data Analysis with NumPy & Pandas	Data Selection & Filtering
19	Module 2: Data Analysis with NumPy & Pandas	Summarizing & Grouping Data
20	Module 2: Data Analysis with NumPy & Pandas	Data Joins & Merges
21	Module 2: Data Analysis with NumPy & Pandas	Advanced Data Transformation
22	Module 2: Data Analysis with NumPy & Pandas	Text Manipulation in DataFrames
23	Module 2: Data Analysis with NumPy & Pandas	The Data Cleaning Workflow
24	Module 2: Data Analysis with NumPy & Pandas	Exploratory Data Analysis (EDA) Concepts
25	Module 2: Data Analysis with NumPy & Pandas	Visualizing Data with Seaborn
26	Module 3: SQL for Data Science	Relational Database Foundations
27	Module 3: SQL for Data Science	Basic SQL Querying
28	Module 3: SQL for Data Science	Data Types & Constraints
29	Module 3: SQL for Data Science	Mastering SQL Joins
30	Module 3: SQL for Data Science	Set Operations & Subqueries
31	Module 3: SQL for Data Science	Relational Integrity & Normalization
32	Module 3: SQL for Data Science	Advanced Aggregations
33	Module 3: SQL for Data Science	Introduction to Window Functions
34	Module 3: SQL for Data Science	Lead; Lag & Time-Series SQL
35	Module 3: SQL for Data Science	SQL + Python Integration
36	Module 3: SQL for Data Science	Database Interactions & Transactions
37	Module 3: SQL for Data Science	Case Study: E-commerce Analysis
38	Module 4: Statistics & Data Visualization	Descriptive Statistics
39	Module 4: Statistics & Data Visualization	Probability Foundations
40	Module 4: Statistics & Data Visualization	Discrete & Continuous Distributions
41	Module 4: Statistics & Data Visualization	Sampling & Estimation
42	Module 4: Statistics & Data Visualization	Hypothesis Testing Basics
43	Module 4: Statistics & Data Visualization	A/B Testing & Correlation
44	Module 4: Statistics & Data Visualization	Principles of Visual Storytelling
45	Module 4: Statistics & Data Visualization	Choosing the Right Chart

46	Module 4: Statistics & Data Visualization	Advanced Seaborn & Interactive Plots
47	Module 4: Statistics & Data Visualization	Introduction to BI Tools
48	Module 4: Statistics & Data Visualization	Building Interactive Dashboards
49	Module 4: Statistics & Data Visualization	Case Study: Executive Reporting
50	Module 5: Applied Machine Learning	Introduction to Machine Learning
51	Module 5: Applied Machine Learning	Linear Regression
52	Module 5: Applied Machine Learning	Logistic Regression
53	Module 5: Applied Machine Learning	Decision Trees & Overfitting
54	Module 5: Applied Machine Learning	Classification Evaluation Metrics
55	Module 5: Applied Machine Learning	K-Nearest Neighbors & Distance Metrics
56	Module 5: Applied Machine Learning	Ensemble Learning: Random Forests
57	Module 5: Applied Machine Learning	Ensemble Learning: Gradient Boosting
58	Module 5: Applied Machine Learning	Unsupervised Learning & Clustering
59	Module 5: Applied Machine Learning	Feature Engineering & Selection
60	Module 5: Applied Machine Learning	Building Robust ML Pipelines
61	Module 5: Applied Machine Learning	Model Debugging & Error Analysis
62	Module 6: GenAI for Data Science	Introduction to GenAI & LLMs
63	Module 6: GenAI for Data Science	Mastering Prompt Engineering
64	Module 6: GenAI for Data Science	Handling Edge Cases & Consistency
65	Module 6: GenAI for Data Science	The OpenAI API & Python Integration
66	Module 6: GenAI for Data Science	Designing Input-Output Contracts
67	Module 6: GenAI for Data Science	Automating Document Workflows
68	Module 6: GenAI for Data Science	Introduction to Embeddings
69	Module 6: GenAI for Data Science	Vector Databases
70	Module 6: GenAI for Data Science	Building Semantic Search
71	Module 6: GenAI for Data Science	RAG: Retrieval-Augmented Generation
72	Module 6: GenAI for Data Science	Advanced RAG: Chunking & Retrieval
73	Module 6: GenAI for Data Science	Building AI Apps with Streamlit
	Capstone Project	Capstone Project: Problem Definition
	Capstone Project	Capstone Project: Pipeline Build
	Capstone Project	Capstone Project: Refinement & Evaluation
	Capstone Project	Capstone Project: Final Dashboard/UI
	Capstone Project	Capstone Project: Documentation & Ethics
	Capstone Project	Final Capstone Showcase